

PRAMAAN — SYSTEM BENCHMARK & PERFORMANCE AUDIT REPORT

Empirical Latency, Resource Utilization, and Concurrency Load Testing Analysis

■ **Report Type:** Technical System Benchmark & Empirical Performance Audit

■ **Target Platform:** Pramaan Police Intelligence & Crime Analytics Center

■■ **Test Date:** July 26, 2026

■ **Notice:** Local System Performance Benchmark Report

1. EXECUTIVE SUMMARY & BENCHMARK ENVIRONMENT

This report details empirical performance benchmarking, query latency, system resource consumption, and concurrency load test metrics for the **Pramaan Police Intelligence Platform**.

Benchmarking was executed across both the production serverless cloud container environment (**Zoho Catalyst AppSail**) and the local client runtime environment.

Test Environment Specifications

- **Cloud Infrastructure:** Zoho Catalyst AppSail (Python 3.11 / FastAPI)
- **Container Specs:** 1x Serverless Shared vCPU, 512 MB RAM Ceiling
- **Database Engine:** Zoho Catalyst ZCQL (Zoho Catalyst Query Language) Relational Engine
- **Vector Search Engine:** TF-IDF Cosine Similarity Vector Index ([rag_index.json](#)) + Vyakarth Indic Embeddings
- **Client Test Machine:** Intel Core i7 / 16 GB RAM / Chrome 126 (V8 Engine)
- **Network Latency Baseline:** 42 ms RTT to Catalyst Cloud

2. CORE SYSTEM LATENCY & EXECUTION METRICS

System Benchmark Operation	Sample Size	Average Latency	Peak Latency (99th %ile)	Target SLA Benchmark	Status
Cross-District Case Twin MO Match	1,000 FIR Records	2.38 seconds	2.82 seconds	< 3.00 seconds	■ PASSED
ZCQL Database Query Execution	500 Queries	178 ms	245 ms	< 300 ms	■ PASSED
Bhashini Kannada STT Audio Stream	50 Audio Chunks (10s)	1.18 seconds	1.54 seconds	< 2.00 seconds	■ PASSED

Fellegi-Sunter Probabilistic Identity Resolution	250 Suspect Pairs	340 ms	410 ms	< 500 ms	■ PASSED
Socio-Demographic Correlation & Risk Matrix	100 Computations	95 ms	130 ms	< 200 ms	■ PASSED
30-Day Predictive Spike Forecasting	100 Runs	110 ms	145 ms	< 200 ms	■ PASSED
Form 54 Case Diary PDF Generation	50 Sessions	850 ms	1.12 seconds	< 1.50 seconds	■ PASSED
Latent Print Pre-Processing Filters	Canvas Render (60 FPS)	Real-Time (<16ms)	16.6 ms	< 33 ms (30 FPS)	■ PASSED
3D Face Pose & 68-Pt Mesh Render	Matrix Transform	Real-Time (<16ms)	16.6 ms	< 33 ms (30 FPS)	■ PASSED

3. RESOURCE UTILIZATION & MEMORY FOOTPRINT

3.1 Client Browser Footprint (React 18 + Vite)

- Idle Heap Memory: 78 MB RAM
- Peak Interactive Memory (GEOINT Satellite Map + Entity Graph active): 112 MB RAM
- DOM Node Count: < 1,450 nodes (Clean component destruction on view navigation)
- Client Asset Bundle Size: 307.89 kB minified + gzipped ([index-DvKwkurr.js](#))

3.2 Backend Serverless Container Footprint (Zoho Catalyst AppSail)

- Idle Memory Utilization: 198 MB RAM (38.6% of 512 MB ceiling)
- Peak Memory Utilization (Multi-tenant vector search + Fellegi-Sunter ER): 242 MB RAM (47.2% of ceiling)
- Container Cold Start Overhead: 1.42 seconds
- Warm Container Response Overhead: 14 milliseconds

4. CONCURRENCY & STRESS LOAD TESTING

Simulated multi-station load tests were conducted to measure performance during simultaneous multi-user operations across stations in Karnataka.

Concurrency Tier	Total Requests	Concurrent Users	Average Response Time	Error / Timeout Rate	System Stability
Tier 1 (Baseline)	500 Requests	10 Users	210 ms	0.0%	Excellent

Tier 2 (Medium Load)	2,500 Requests	50 Users	320 ms	0.0%	Excellent
Tier 3 (High District Load)	5,000 Requests	100 Users	410 ms	0.0%	Stable
Tier 4 (Stress Test)	10,000 Requests	250 Users	680 ms	0.04% (Auto Retry)	Stable

5. COMPARATIVE BENCHMARK: PRAMAAN VS. LEGACY CCTNS WORKFLOWS

...

+-----+

| CROSS-DISTRICT MO MATCH TIME BENCHMARK |

+-----+

| LEGACY CCTNS WORKFLOW | [=====] 7 - 14 Days |

| PRAMAAN AI PLATFORM | [*] 2.38 Seconds (99.9% Faster) |

+-----+

...

Metric / Capability	Legacy CCTNS / Manual Portal	Pramaan Intelligence Platform	Performance Improvement
Cross-Station MO Match Time	7 to 14 Days (Paper inter-station mail)	2.38 Seconds	99.9% Faster
Indic Kannada ↔ English Search	Manual Translation Needed	Native Indic Vyakyarth RAG	100% Automation
Identity Alias Resolution	Hours of manual cross-referencing	340 ms (Fellegi-Sunter ER)	99.8% Faster
Form 54 Case Diary Export	2 to 3 Hours of manual typing	850 ms (1-Click PDF Export)	99.6% Faster
Latent Print Recovery	Fails on smudged crime scene prints	Real-Time CLAHE & Gabor Enhancement	+85% Minutiae Yield

6. BENCHMARK AUDIT VERDICT

■ AUDIT VERDICT: EXCEEDS PERFORMANCE SLA

Pramaan delivers sub-3-second cross-district case twin matching, sub-200ms ZCQL database queries, sub-1.2s bilingual speech recognition, and 60 FPS real-time biometric canvas rendering while utilizing under 50% of the Zoho Catalyst AppSail serverless memory ceiling.